Laboratory Record

of COMPUTER NETWORKS LAB

Roll No.: 1601-24-733-

Experiment No.: Lab Task - 11

Sheet No.:

Date.:

LAB TASK – 11

Aim: Implement Socket Programming.

Description:

In this we have to implement a TCP client-server program in the C programming language. Here, the client and server would exchange messages and communicate with each other in an interactive way. In addition, you will also learn about the client-server architecture and other related concepts.

A **socket** is a structure that allows communication between different processes on the same computer or different computers. A socket allows us to communicate by sending and receiving data over the network. In simple terms, it is a way for a computer to talk to other computers on the network.

A **client-server architecture** is a model in computer networking, where the server provides some service to the client. In this architecture, the client computer sends a request to the server computer through a network (internet). The server accepts this request and sends the required data to the client.
In simple terms, we can say :

- Server is a remote computer which provides some services.
- Client is the one that request the server for these services.

CODE :

//tcp_server_chat.py

```
import socket
server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server_ip = "0.0.0.0"
port = 12345
server_socket.bind((server_ip, port))
server_socket.listen(1)
print(f"Server is listening on port {port}...")
conn, addr = server_socket.accept()
print(f"Connected with client {addr}")
while True:
    msg = conn.recv(1024).decode()
    if msg.lower() == "bye":
        print("Client ended chat.")
        break
    print(f"Client: {msg}")
    reply = input("Server: ")
    conn.send(reply.encode())
    if reply.lower() == "bye":
        print("Server ended chat.")
        break
conn.close()
server_socket.close()
```

//tcp_client_chat.py

```
import socket
client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server_ip = input("Enter Server IP Address: ")
port = 12345
client_socket.connect((server_ip, port))
print(f"Connected to server {server_ip}:{port}")
while True:
    msg = input("Client: ")
    client_socket.send(msg.encode())

    if msg.lower() == "bye":

        print("Client ended chat.")
        break
    reply = client_socket.recv(1024).decode()
    if reply.lower() == "bye":
        print("Server ended chat.")
        break
    print(f"Server: {reply}")
client_socket.close()
```

//udp_server_chat.py

```
import socket
server_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
server_ip = "0.0.0.0"
port = 12345
server_socket.bind((server_ip, port))
print(f"UDP Server is listening on port {port}...")
while True:
    msg, client_addr = server_socket.recvfrom(1024)
    msg = msg.decode()
    print(f"Connected with client {client_addr}")
    if msg.lower() == "bye":
        print("Client ended chat.")
        break
    print(f"Client: {msg}")
    reply = input("Server: ")
    server_socket.sendto(reply.encode(), client_addr)
    if reply.lower() == "bye":
        print("Server ended chat.")
        break
server_socket.close()
```

//udp_client_chat.py

```
import socket
client_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
server_ip = input("Enter Server IP Address: ")
port = 12345
print(f"Connected to UDP server at {server_ip}:{port}")
while True:
    msg = input("Client: ")
    client_socket.sendto(msg.encode(), (server_ip, port))
    if msg.lower() == "bye":
print("Client ended chat.")
break
data, server_addr = client_socket.recvfrom(1024)
reply = data.decode()
print(f"Server: {reply}")
if reply.lower() == "bye":
print("Server ended chat.")
break
client_socket.close()
```

```
PS C:\Users\CSELAB3-28\Desktop> python .\tcp_server_chat.py
Server is listening on port 12345...
Connected with client ('127.0.0.1', 55225)
Client: hello
Server: hai
```

```
PS C:\Users\CSELAB3-28> cd .\Desktop\
PS C:\Users\CSELAB3-28\Desktop> python .\tcp_client_chat.py
Enter Server IP Address: 127.0.0.1
Connected to server 127.0.0.1:12345
Client: hello
Server: hai
```

```
C:\Users\CSELAB3-28\Desktop>python udp_server_chat.py
UDP Server is listening on port 12345...
Connected with client ('127.0.0.1', 52981)
Client: Hello
Server: Hai
Connected with client ('127.0.0.1', 52981)
Client ended chat.

C:\Users\CSELAB3-28\Desktop>bye
```

```
C:\Users\CSELAB3-28\Desktop>python udp_client_chat.py
Enter Server IP Address: 127.0.0.1
Connected to UDP server at 127.0.0.1:12345
Client: Hello
Server: Hai
Client: bye
Client ended chat.

C:\Users\CSELAB3-28\Desktop>
```